

Aggregate functions:
Name: Sagi Sahasra Srihasa
Roll no : 2520030008

Step 1:

```
1
2  /* Name: Sagi Sahasra Srihasa
3     Roll no : 2520030008
4     */
5
6
7  -- Step 1: Create Database and Table
8  • CREATE DATABASE aggregate_bank_db;
9  • USE aggregate_bank_db;
10 • CREATE TABLE bank_transactions (
11     txn_id INT PRIMARY KEY,
12     customer_name VARCHAR(50),
13     branch_name VARCHAR(50),
14     transaction_type VARCHAR(20),
15     amount DECIMAL(10,2),
16     transaction_date DATE
17 );
```

Output

#	Time	Action	Message
✓ 1	21:27:01	CREATE DATABASE aggregate_bank_db	1 row(s) affected
✓ 2	21:27:03	USE aggregate_bank_db	0 row(s) affected
✓ 3	21:27:06	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected

Step 2:

```
-- Step 2: Insert Data
• INSERT INTO bank_transactions VALUES
(101, 'Sahasra', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
(102, 'Tejaswini', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
(103, 'Manu', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
(104, 'Latha', 'Vizag', 'Deposit', 8000, '2024-01-10'),
(105, 'Ravi', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
(106, 'Pranitha', 'Vizag', 'Deposit', 15000, '2024-01-12'),
(107, 'Angela', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
(108, 'Bhanu Priya', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
(109, 'Nikshitha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
(110, 'Keerthi', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
```

✓ 4 21:36:51 INSERT INTO bank_transactions VALUES (101,'Sahasra','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Tejaswi...', 10 row(s)

Step 3:

```
30 -- Step 3: SUM()
31 • SELECT SUM(amount) AS Total_Amount
32 FROM bank_transactions;
```

Result Grid	Filter Rows:	Export:	Wrap
Total_Amount			
70500.00			

✓ 5 21:37:29 SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000

Step 4:

```
33 -- Step 4: AVG()
34 • SELECT AVG(amount) AS Average_Trans
35 FROM bank_transactions;
```

Result Grid	Filter Rows:	Exp
Average_Transaction		
7050.000000		

✓ 6 21:38:24 SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000

Step 5:

```
37 • SELECT MAX(amount) AS Highest_Transaction
38 FROM bank_transactions;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Highest_Transaction			
15000.00			

✓ 7 21:39:42 SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000

Step 6:

```
40 • SELECT MIN(amount) AS Lowest_Transaction
41 FROM bank_transactions;
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	Lowest_Transaction
▶	1000.00

✓ 8 21:40:26 SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000 1 row(s) returned

Step 7:

```
43 • SELECT COUNT(*) AS Total_Transactions
44 FROM bank_transactions;
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	Total_Transactions
▶	10

✓ 9 21:41:07 SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000 1 row(s) returned

Step 8:

```
47 -- Step 8: WHERE + SUM()
48 • SELECT SUM(amount) AS Total_Deposit
49 FROM bank_transactions
50 WHERE transaction_type = 'Deposit';
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	Total_Deposit
▶	60000.00

✓ 10 21:41:58 SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000 1 row(s) returned

Step 9:

```
53      -- Step 9: GROUP BY + SUM()
54  •    SELECT branch_name,
55          SUM(amount) AS Total_Amount
56  FROM bank_transactions
57  GROUP BY branch_name;
58
59
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	branch_name	Total_Amount			
▶	Hyderabad	19500.00			
	Vijayawada	24000.00			
	Vizag	27000.00			

Step 10:

```
60      -- Step 10: GROUP BY + HAVING
61  •    SELECT branch_name,
62          SUM(amount) AS Total_Amount
63  FROM bank_transactions
64  GROUP BY branch_name
65  HAVING SUM(amount) > 20000;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	branch_name	Total_Amount			
▶	Vijayawada	24000.00			
	Vizag	27000.00			

✓ 12 21:43:54 SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n... 2 row(s) returned

Step 11:

```
66 -- Step 11: GROUP BY + ORDER BY
67 • SELECT branch_name,
68       SUM(amount) AS Total_Amount
69 FROM bank_transactions
70 GROUP BY branch_name
71 ORDER BY Total_Amount DESC;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	branch_name	Total_Amount			
▶	Vizag	27000.00			
	Vijayawada	24000.00			
	Hyderabad	19500.00			

✓ 13 21:45:11 SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n... 3 row(s) returned

Step 12:

```
72 -- Step 12: GROUP BY + HAVING + ORDER BY
73 • SELECT branch_name,
74       COUNT(*) AS Total_Transactions
75 FROM bank_transactions
76 GROUP BY branch_name
77 HAVING COUNT(*) >= 3
78 ORDER BY Total_Transactions DESC;
79
80
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	branch_name	Total_Transactions			
▶	Hyderabad	4			
	Vijayawada	3			
	Vizag	3			

Step 13:

```
81 -- Step 13: WHERE + COUNT()
82 • SELECT COUNT(*) AS Withdrawals
83 FROM bank_transactions
84 WHERE transaction_type = 'Withdrawal';
85
86
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Withdrawals			
4			

Step 14:

```
87 -- Step 14: WHERE + AVG()
88 • SELECT AVG(amount) AS Avg_Deposit
89 FROM bank_transactions
90 WHERE transaction_type = 'Deposit';
91
92
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Avg_Deposit			
10000.000000			

Step 15:

```
93 -- Step 15: GROUP BY + MAX()
94 • SELECT branch_name,
95        MAX(amount) AS Highest_Amount
96 FROM bank_transactions
97 GROUP BY branch_name;
98
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Highest_Amount		
Hyderabad	9000.00		
Vijayawada	12000.00		
Vizag	15000.00		

Step 16:

```
99      -- Step 16: GROUP BY + MIN()
100 •    SELECT branch_name,
101           MIN(amount) AS Lowest_Amount
102 FROM bank_transactions
103 GROUP BY branch_name;
104
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Lowest_Amount		
Hyderabad	2000.00		
Vijayawada	1000.00		
Vizag	4000.00		

Step 17:

```
105      -- Step 17: WHERE + ORDER BY
106 •    SELECT *
107 FROM bank_transactions
108 WHERE amount > 8000
109 ORDER BY amount DESC;
110
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	Pranitha	Vizag	Deposit	15000.00	2024-01-12
	103	Manu	Vijayawada	Deposit	12000.00	2024-01-08
	110	Keerthi	Vijayawada	Deposit	11000.00	2024-01-16
	108	Bhanu Priya	Hyderabad	Deposit	9000.00	2024-01-14
✱	NULL	NULL	NULL	NULL	NULL	NULL

Step 18:

```
111 -- Step 18: GROUP BY + COUNT()
112 • SELECT branch_name,
113        COUNT(*) AS Customer_Count
114 FROM bank_transactions
115 GROUP BY branch_name;
116
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Customer_Count		
Hyderabad	4		
Vijayawada	3		
Vizag	3		

Step 19:

```
117 -- Step 19: HAVING + COUNT()
118 • SELECT branch_name,
119        COUNT(*) AS Total_Transactions
120 FROM bank_transactions
121 GROUP BY branch_name
122 HAVING COUNT(*) > 3;
123
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
branch_name	Total_Transactions		
Hyderabad	4		

Step 20:

```
127 -- Step 20: GROUP BY + AVG()
128 • SELECT branch_name,
129        AVG(amount) AS Avg_Amount
130 FROM bank_transactions
131 GROUP BY branch_name;
132
133
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	branch_name	Avg_Amount
▶	Hyderabad	4875.000000
	Vijayawada	8000.000000
	Vizag	9000.000000

Step 21:

```
134 -- Step 21: WHERE + GROUP BY + HAVING
135 • SELECT branch_name,
136        SUM(amount) AS Total_Deposit
137 FROM bank_transactions
138 WHERE transaction_type = 'Deposit'
139 GROUP BY branch_name
140 HAVING SUM(amount) > 15000;
141
142
143
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	branch_name	Total_Deposit
▶	Vijayawada	23000.00
	Vizag	23000.00

Step 22:

```
144 -- Step 22: GROUP BY + HAVING + ORDER BY
145 • SELECT branch_name,
146        AVG(amount) AS Avg_Amount
147 FROM bank_transactions
148 GROUP BY branch_name
149 HAVING AVG(amount) > 7000
150 ORDER BY Avg_Amount DESC;
```

Result Grid		 Filter Rows:	Export: 	Wrap Cell Content: 
	branch_name	Avg_Amount		
▶	Vizag	9000.000000		
	Vijayawada	8000.000000		

✓ 24 21:50:18 SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_na... 2 row(s) returned